

GENERAL OSTEOLOGY

PLAN

1. INTRODUCTION

2. STRUCTURE OF THE SKELETON

3. TYPES OF BONES

4. BONE MORPHOLOGY

Objectives:

- Recognize the components of the skeleton
- Classify bones according to their shape
- Understand bone morphology

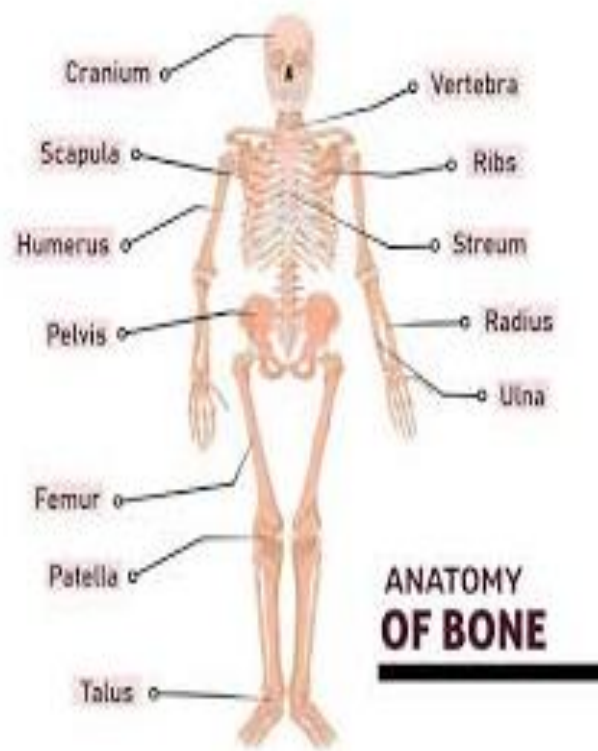
1. INTRODUCTION:

OSTEOLOGY is the science that studies bones.

The skeletal system is the set of bones (white and hard organs) and cartilages that serve to:

- Support: provides a framework on which muscles are arranged
- Protect organs
- Allow movement
- Store calcium and phosphorus
- Hematopoiesis (formation of blood cells by the bone marrow)

The skeletal system is made up of 206 constant bones and other variable bones represented by the Wormian bones of the skull and the sesamoid bones located in tendons.



SKELETON

2. STRUCTURE OF THE SKELETON:

The skeleton is composed of an axial skeleton to which the appendicular skeleton is attached.



2.1 Axial skeleton:

It includes the bones of the head, the hyoid bone, the vertebral column, and the thoracic skeleton.

- ❖ Bones of the head: grouped into the bones of the skull and the bones of the face.

- * The fused skull bones form the cranial cavity that contains part of the central nervous system and the initial portion of the cranial nerves.

- * The fused facial bones delimit cavities that house the sense organs, and the initial parts of the respiratory and digestive systems.

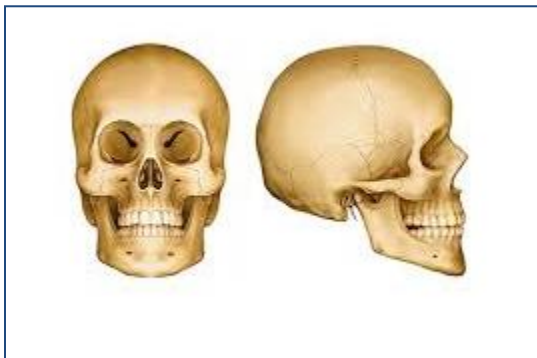
- ❖ Hyoid bone: located at the upper limit of the neck, at a distance from the other skeletal bones.

- ❖ Vertebral column: hollowed by a canal containing the spinal cord and spinal nerve roots, it consists of 33 vertebrae, divided into:

- * The mobile column: 7 cervical vertebrae, 12 thoracic vertebrae, 5 lumbar vertebrae

- * The fixed column: vertebrae fused into two bones – the sacrum and the coccyx

- ❖ Thoracic skeleton: made up of 12 pairs of ribs and the sternum.



BONES OF THE HEAD



HYOID BONE

- ❖ Lower limbs (pelvic limbs): attached to the vertebral column by the pelvic girdle (two hip bones).

The lower limb includes: femur, tibia, fibula, tarsal bones, metatarsals, and phalanges.



LOWER LIMBS

3. **TYPES OF BONES:**

According to their shape, bones are classified into 4 types:

3.1 Long bones:

Bones where length predominates, with two extremities (epiphyses), an intermediate part (shaft or diaphysis), and a junction between them called metaphysis.

Examples: humerus, femur, ulna.

3.2 Flat bones:

Their thickness is clearly less than their other dimensions. They have two surfaces, borders, and angles.

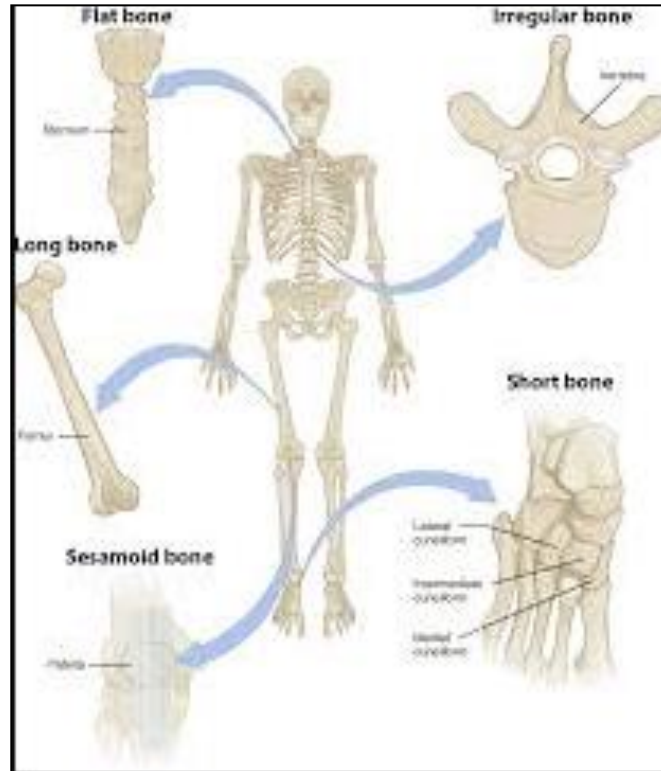
Examples: scapula, parietal bone.

3.3 Short bones:

They have three almost equal dimensions, Example: carpal bones.

3.4 Irregular bones:

Examples: vertebrae (with several processes), pneumatic bones (hollowed with a cavity, like the maxilla), etc.



4. BONE MORPHOLOGY:

The surface of bones is often irregular, showing projections or depressions, which may be articular or non-articular.

4.1 Bone projections (eminences):

- Articular projections:

- * Head: a projection clearly separated from the bone by a narrowed area (neck)
- * Condyle: a prominent articular surface

- Non-articular projections: usually serve as points for muscle force application or ligament attachment.

Depending on size:

- * Process (apophysis): large bony outgrowth, e.g., coracoid process
- * Tuberosity: broad projection, less detached, e.g., ischial tuberosity
- * Tubercle: small rounded eminence, e.g., infraglenoid tubercle of the scapula

* Spine: sharp, pointed projection, e.g., ischial spine

* Crest: linear projection, e.g., iliac crest

Note: Some bony prominences are subcutaneous and therefore palpable, serving as clinical landmarks.

4.2 Bone depressions (cavities):

- Articular cavities: correspond to articular projections, e.g., glenoid cavity of the scapula.

- Non-articular cavities: serve for the passage or attachment of anatomical structures:

* Groove (sulcus): an open channel, e.g., radial groove

* Fossa: a shallow or deep depression, e.g., olecranon fossa

* Notch (incisure): indentation on a border, e.g., ischial notch

* Foramen: opening in a bone, e.g., obturator foramen



References:

1. Kamina P. Clinical Anatomy, Volume 1: General Anatomy, Limbs. 4th edition, Maloine.(traduit en anglais)
2. Rouvière H. Human Anatomy: Descriptive and Topographic. Volume 1: Head and Neck. Masson, 10th edition, 2002.(traduit en anglais).